PSE-Net: Relational Symbolic Data-Link Framing Protocol

Author: M. K. Almodiny
DOI: 10.5281/zenodo.23846190
Ref: Reference

Reference Concept Whitepaper, System Architecture, & Standard Specification

1. EXECUTIVE SUMMARY & VISION

The **PSE-Net (Weighted Pair-Symbol Encoding Network Protocol)** introduces a novel algorithmic and structural framework for data framing across Data-Link and Transport layers. Built around a stateful *Within-State Ladder Rule* with BCD frame alignment and CRC-32 structural integrity validation, it delivers dependable binary payload transport without requiring heavy third-party dependencies.

0%	16-Byte	C99 / Py	CRC-32
External Dependencies	Standard Header Size	Dual-Core Implementations	Integrity Validation

2. ARCHITECTURE & SYMBOL ENCODING

The core mechanism transforms raw binary byte streams into weighted symbolic representations bounded by dynamic states. The protocol decouples framing into two primary stages:

> **Within-State Ladder Rule:** The algorithm analyzes sequential binary pairs and applies ladder state transitions ($S_{n+1} = f(S_n, P_i)$) to minimize repetitive symbol transitions and enforce structural signal stability. The current ladder state is computed dynamically from pair weights with continuous signal balance checks.

3. FRAME HEADER SPECIFICATION

The header is engineered for ultra-low latency and zero memory overhead, packing critical framing metadata into a compact 16-byte fixed structure:

Offset	Field Name	Size	Technical Description
`0x00 - 0x01`	Magic Identifier	2 Bytes	Protocol magic constant (`0x5053` "PS") for fast frame boundaries
`0x02 - 0x03`	Sequence Number	2 Bytes	Monotonic sequence tracker for packet loss detection
`0x04 - 0x07`	Payload Length	4 Bytes	Length of encoded payload binary in bytes (Big-Endian format)
`0x08 - 0x0B`	Ladder State Flags	4 Bytes	Active symbolic ladder state flags and state transition parameters
`End - 4B`	CRC-32 Checksum	4 Bytes	Standard Ethernet CRC-32 polynomial (`0xEDB88320`) for frame verification

4. PERFORMANCE BENCHMARKING & COMPARATIVE ANALYSIS

The following comparative matrix evaluates PSE-Net against standard framing and serialization schemes across key metrics such as overhead, memory footprint, and embedded suitability:

Protocol / Standard	Overhead Ratio	Memory Footprint	Embedded Support (C)	Primary Use Case
PSE-Net Plain	Low (Fixed + 16B)	Zero-Allocation (Stack/Static)	Excellent (C99 Header-only)	Error-resilient binary transport & framing
Base64 Encoding	High (+33.3%)	Low to Moderate	Supported	Text-based MIME/Web payload encoding
JSON over TCP	Very High (+100-300%)	Heavy (DOM/Parser Required)	Poor / Unsuitable	Top-layer Web API application data exchange
Protocol Buffers	Ultra-compact	Moderate (Requires Code Generator)	Supported (via nanopb)	High-performance inter-service RPC (gRPC)

5. SECURITY MODEL & EVOLUTIONARY ROADMAP

PSE-Net adheres strictly to the separation of concerns principle, delegating cryptographic security and identity verification to robust transport-layer standards:

- **Security Layering:** Fully compatible as encapsulated payload over **TLS 1.3**, **Noise Protocol Framework**, or **IPsec** tunnels.
- **C99 Embedded Kernel:** Single-header C99 implementation featuring zero dynamic memory allocation (`malloc-free`) for real-time microcontrollers.
- **Standardization Path:** Drafting the formal IETF RFC-style specification to support independent client/server decoders in Go, Rust, and C++.

6. PROTOCOL LIMITATIONS & OUT-OF-SCOPE FEATURES

The PSE-Net Plain protocol provides a dedicated reference implementation focused strictly on the framing and symbolic encoding layer (Data Link framing). It does not replace full-featured transport, session, or security protocol stacks. The following capabilities fall outside the scope of the current implementation:

1. Transport & Flow Control

- **No Retry / Retransmission:** Lacks automated mechanisms to request retransmission of lost or dropped frames upon delivery failure.
- **No Sequence Numbering:** Does not track or enforce packet sequence numbers at the protocol framing layer.
- **No ACK / Smart Retransmission:** Omits interactive acknowledgment (ACK) signaling for intelligent loss recovery.
- **No Congestion Control:** Contains no dynamic flow-control mechanisms to throttle transmission rates based on network load or congestion.
- **No Professional Fragmentation / MTU Awareness:** Does not fragment or reassemble payloads according to network Maximum Transmission Unit (MTU) boundaries.

2. Session & Concurrency

- **No Complete Network State Machine:** Omits complex network state machines required to handle real-world network disconnections, reconnects, or automated fault recovery.
- **No Concurrent Clients / Multi-Streaming:** The reference implementation is not architected to serve concurrent client connections or multiplexed streams in production environments.
- **No Session Management & Persistence:** Session store mechanisms, state persistence, and session resumption after drops are unsupported.
- **Limited Handling for Out-of-Order Frames:** Damaged or out-of-order incoming socket frames are dropped without complex frame reordering.

3. Security Context & Secure Transport

- **No Authentication or Security Context:** The standard protocol header includes no native cryptographic identity verification or inline encryption.
- **External Secure Transport Requirement:** PSE-Net relies on external infrastructure-level protection, including:
  - **TLS** (Transport Layer Security) tunnels.
  - **SSH Tunnels**.
  - Separate payload-level encryption (e.g., **AES-GCM**).
  - Deployment inside isolated private networks (**VPN / Private LAN**).